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## **Accurate Lithium Quantification in Hypersaline Brines Using a Polybutadiene Maleic Acid Based Ion Exchange Column**

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### **Abstract**

Accurately measuring lithium content in hypersaline brines with total dissolved solids (TDS) exceeding 150,000 ppm is challenging due to matrix interferences and dilute lithium concentrations. The objective of this work is to demonstrate a reliable method based on polybutadiene maleic acid (PBMA) resin for precise lithium quantification, even at low concentrations and in highly complex brine matrices, reducing the uncertainty in prospecting for lithium brine resources.

An ion chromatography system with a degasser, suppressor, ion exchange columns, and conductivity detectors was used. Samples were diluted to keep TDS below 500 ppm to protect the ion exchange columns. Suspended solids were removed using 0.45-micron syringe filters. The cation-exchange column was based on a PBMA resin. The method's accuracy was evaluated using seven synthetic brines containing lithium, sodium, potassium, magnesium, calcium, strontium, barium, chloride, bromide, and iodide. Lithium concentrations were calculated from conductivity-based chromatograms through numerical integration, and density measurements were used for mg/L to ppm conversions.

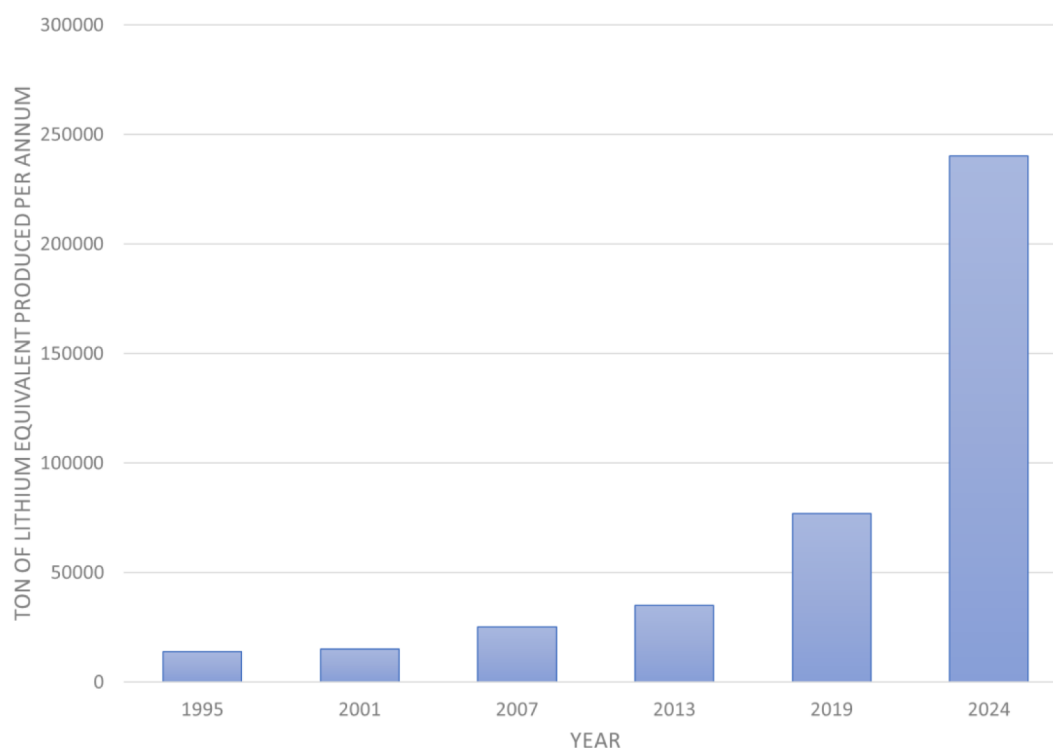
Lithium calibration using standard solutions yielded a linear regression coefficient of 0.99998 with a relative standard deviation of 0.583%. The lithium peak was numerically integrated from chromatograms based on conductivity measurements ( $\mu\text{S}/\text{cm}$  vs. time in minutes). Among the detected cations, lithium exhibited the shortest retention time (6-6.5 minutes), potentially reducing matrix effects and minimizing peak overlap. This behavior is attributed to the ion exchange resin's selectivity and lithium's distinct physicochemical properties. Test brines with TDS exceeding 220,000 ppm were analyzed, containing lithium concentrations between 10 and 200 ppm. The correlation coefficient obtained from the measured brine samples was 0.9983, suggesting that the method effectively quantified lithium with limited matrix interference. Compared to conventional spectroscopic methods, this approach demonstrated promising performance under extreme salinity conditions. This can enable the accurate prospecting of brine resources for lithium.

This study introduces a robust method for lithium quantification in brines with significant matrix interferences. While typical analytical methods for diluting lithium solutions using spectroscopic or spectrometric techniques, this work demonstrates the reliability of PBMA resins in ion chromatography,

enabling precise lithium measurements with simple conductivity detection. This approach can support lithium exploration and prospecting efforts by offering a reliable analytical method for complex brine systems.

## Introduction

Lithium is a critical energy transition mineral owing to its pivotal role in energy storage. Its unique properties make it an irreplaceable component of lithium-ion batteries (LIBs) which are an essential component of electric vehicles (EVs) and an important driver for the energy transition. Carbon emissions is one of the major environmental challenges that need to be addressed through carbon capture utilization and storage (CCUS) solutions such as injecting CO<sub>2</sub> into deep geological formations (Alshammari et al. 2025). Nevertheless, geological storage is difficult due to its high cost and the complex multiphase interactions involved in porous geological formations (Alshammari et al. 2024a). As the world transitions to renewable energy sources and strives to reduce carbon emissions, the importance of lithium is poised to escalate, driven by the growing demand for EVs, solar and wind power systems, and other applications that rely heavily on efficient energy storage solutions. The demand for LIBs has experienced a significant surge over the past decade, with current raw material production projected to be insufficient to meet demand by 2040 (Maisel et al. 2023). This growing demand has driven a substantial increase in global lithium production, rising from 14,000 lithium equivalent tons per annum in 1995 to 240,000 in 2024, as illustrated in Figure 1. According to the 2025 United States Geological Survey (USGS) mineral commodity summaries (National Minerals Information Center, 2025), Australia is the top lithium producer, primarily through hard-rock mining, followed by Chile and China, with the production and reserves of lithium by country detailed in Table 1. The primary application of lithium is in the manufacturing of LIBs, which are crucial for the widespread adoption of renewable energy technologies and electric vehicles. However, lithium also has other notable applications, including the production of ceramics and glass (Lohbauer et al. 2024), lubricating greases (Lugt 2016), and various other industrial uses. Traditional lithium extraction methods, including evaporation from brine sources and hard rock mining, are characterized by their resource-intensive and time-consuming nature (Vera et al. 2023). In contrast, emerging techniques such as Direct Lithium Extraction (DLE) have shown promise in enhancing recovery rates while minimizing environmental impact, leveraging selective extraction mechanisms like adsorption, ion exchange, solvent extraction, and electrochemical processes (Farahbakhsh et al. 2024). The current exploitation of natural lithium resources encompasses three primary sources: (1) salar brines, predominantly found in South America's lithium triangle (Steinmetz and Salvi 2021), (2) hard rock mining of spodumene (Dessemond et al. 2019) and pegmatite (Karrech et al. 2020) deposits, and (3) dilute brines from salt lakes (Song et al. 2017), geothermal (Stringfellow and Dobson 2021), and oilfield brines (Knapik et al. 2023). The advent of advanced DLE technologies has enabled the efficient recovery of lithium from these sources, even at low concentrations, with high selectivity and numerous advantages over traditional methods, including higher recovery rates, reduced land use, and accelerated production cycles (Farahbakhsh et al. 2024). Notably, oilfield brines represent a promising lithium source, given their vast annual production volumes worldwide. However, the low lithium concentrations in these brines, typically ranging from 10 to 200 ppm depending on the geological formation (Kumar et al. 2019), pose significant measurement challenges, particularly for Produced Water (PW) with complex chemistry characterized by high total dissolved solids (TDS) exceeding 150,000 mg/L, dissolved H<sub>2</sub>S, suspended solids, and dispersed hydrocarbon contaminants. As a result, accurate lithium measurement and extraction from oilfield brines require innovative solutions to overcome these technical hurdles.



**Figure 1—Lithium production from 1995 until 2024 (U.S. Geological Survey Mineral Commodity Summaries 1996; U.S. Geological Survey Mineral Commodity Summaries 2002; U.S. Geological Survey Mineral Commodity Summaries 2008; National Minerals Information Center 2014; National Minerals Information Center 2020; National Minerals Information Center 2025).**

**Table 1—Global lithium production and reserves according to the USGS 2025 Mineral Commodity Summaries Report (National Minerals Information Center, 2025).**

| Country               | Mine Production 2023 | Mine Production 2024 | Reserves   |
|-----------------------|----------------------|----------------------|------------|
| United States         | Withheld             | Withheld             | 1,800,000  |
| Argentina             | 8,630                | 18,000               | 4,000,000  |
| Australia             | 91,700               | 88,000               | 7,000,000  |
| Brazil                | 5,260                | 10,000               | 390,000    |
| Canada                | 3,240                | 4,300                | 1,200,000  |
| Chile                 | 41,400               | 49,000               | 9,300,000  |
| China                 | 35,700               | 41,000               | 3,000,000  |
| Namibia               | 2,700                | 2,700                | 14,000     |
| Portugal              | 380                  | 380                  | 60,000     |
| Zimbabwe              | 14,900               | 22,000               | 480,000    |
| Other countries       | -                    | -                    | 2,800,000  |
| Would total (rounded) | 204,000              | 240,000              | 30,000,000 |

In the analysis of lithium concentrations, one of the most common techniques include Inductively Coupled Plasma-Atomic Emission Spectroscopy (ICP-AES), Inductively Coupled Plasma-Optical Emission Spectroscopy (ICP-OES) and Inductively Coupled Plasma-Mass Spectrometry (ICP-MS). They differ in sensitivity, detection limits, argon gas consumption, susceptibility to interferences, operational complexity and applications suitability. ICP-AES and ICP-OES are often used interchangeably, as both rely on detecting light emitted by excited atoms in a plasma. These methods offer detection limits in the low parts per billion (ppb) range, consume relatively low argon gas and are comparatively easy to operate.

However, they struggle to accurately quantify lithium in complex matrices due to potentially significant interference effects. ICP-MS provides much higher sensitivity with detection limits reaching to parts per trillion (ppt) range. This makes it ideal for trace-level lithium detection in complex matrices such as biological or environmental samples. While ICP-MS offers superior precision and accuracy, it has higher argon gas consumption and it is more expensive and require rigorous sample preparation and calibration techniques. The choice among these techniques depends on the required detection limits, sample complexity and analytical budget.

Lithium detection using other spectroscopic methods such as Atomic Absorption Spectroscopy (AAS), Laser-Induced Breakdown Spectroscopy (LIBS), and Flame Emission Spectroscopy (FES) offers varying levels of sensitivity, cost, and applicability depending on the sample matrix and detection requirements. AAS is a well-established technique that quantifies lithium based on its absorption of light at a specific wavelength (670.8 nm), offering detection limits in the low ppm range and good selectivity, especially for aqueous samples and quality control in industrial processes. It is relatively simple and cost-effective but limited to single-element analysis and requires careful calibration. LIBS, on the other hand, uses a high-energy laser pulse to ablate the sample and form a plasma, analyzing the emitted light spectrum to detect lithium. LIBS is highly versatile, enabling rapid, in-situ analysis of solids, liquids, and even aerosols without the need for extensive sample preparation. However, it offers moderate sensitivity and is more prone to matrix effects and signal variability. FES is one of the earliest techniques used for lithium detection and relies on measuring the light emitted by lithium atoms when excited in a flame. While it is straightforward and low-cost, FES suffers from low selectivity and accuracy and has largely been replaced by more advanced techniques such as AAS and ICP-based methods. Nevertheless, all three techniques remain relevant for specific applications, particularly where rapid or field-based lithium screening is required.

IC is considered a simple and relatively low-cost technique for ions detection due to its straightforward operation, minimal sample preparation and the affordability of equipment compared to advanced spectroscopic methods like ICP-MS separate lithium ions based on their interaction with an ion-exchange resin and detect them using conductivity or suppressed conductivity detectors, which are less expensive and easier to maintain than plasma-based analytical systems. Additionally, IC systems do not require expensive gasses like argon or high temperature plasma sources, reducing both operational and maintenance costs. The method is well suited for routine analysis of lithium in water, brines and industrial effluents where detection limits in low ppm to sub-ppm range are acceptable. Its ease of use, lower training requirement and cost effectiveness make it a regular choice in environmental monitoring and quality control laboratories. In this study our focus is on PW which is a byproduct of oil and gas extraction, typically exhibits high salinity and complex ionic composition, reflecting interactions between formation water, injected fluids, and reservoir rock (Alshammari 2024b). Chemically, it is often characterized by elevated concentrations of major cations such as sodium ( $\text{Na}^+$ ) and calcium ( $\text{Ca}^{2+}$ ), and anions like chloride ( $\text{Cl}^-$ ), which dominate the ionic balance (Collins 1975). Sodium concentrations can range from several thousand to over 100,000 mg/L, while calcium levels are often substantial due to dissolution of carbonate and evaporite minerals. Chloride is usually the most abundant anion, contributing significantly to the total dissolved solids (TDS), which can exceed 200,000 mg/L in some formations. In contrast, lithium ( $\text{Li}^+$ ) is typically present in trace amounts, often below 10 mg/L, due to its lower natural abundance and limited geochemical mobility under typical reservoir conditions (Collins 1975). The high ionic strength and dominance of interfering ions such as  $\text{Na}^+$  and  $\text{Ca}^{2+}$  pose challenges for lithium detection and extraction, as they can outcompete lithium in ion-exchange and separation processes. This complex matrix requires careful analytical and process design, especially when evaluating the potential for lithium recovery from PW as a secondary resource. Despite the importance of lithium measurements to quantify the potential lithium brine resources in PW, to the best of our knowledge, a detailed analysis for methods to measure the low concentration of lithium in PW are limited in the literature. In this work, we detailing the procedure and accuracy of using IC based method with conductivity detection using PBMA column and this work bridge this gap by providing

experimental evidence for the accuracy of measuring lithium in high TDS synthetic PW samples with a varying concentration of lithium.

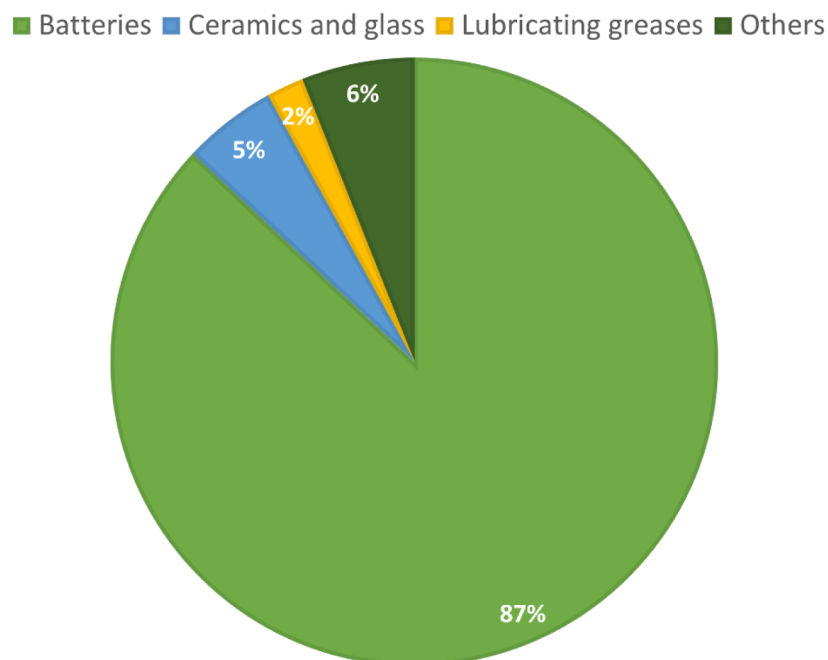


Figure 2—Estimates of global lithium end uses (National Minerals Information Center, 2025)

## Methods

For sample preparation rigorous sample preparation was done of synthetic PW, which typically contains over 150,000 ppm of total dissolved solids (TDS). Synthetic PW brines were formulated based on the chemical components listed in Table 2 and compositions in Table 3. To ensure that ion concentrations fell in range of the IC calibration curve data points 0.05, 0.1, 0.15, 0.2, 1, 5 and 10 ppm and to protect the PBMA column from salt overloading, the samples were diluted 1000 times by transferring 100  $\mu\text{L}$  of the synthetic PW using (100  $\mu\text{L}$  eppendorf pipette) and filled to the mark with 18.2 M $\Omega$ -cm pure water in 100 mL volumetric flask. This dilution not only enabled reliable quantification of ions but also prolonged the lifetime of the PBMA resin-coated silica column. Prior to analysis, samples were filtered using 0.45  $\mu\text{m}$  syringe membrane filters to remove particulates and prevent clogging or baseline drift. After that TDS measurement were performed using TDS Meter (Myron L 532T1) pre- and post-dilution to validate dilution accuracy and assess the ionic strength of the matrix. IC eluents were prepared (nitric acid 69%, Sigma-Aldrich 0.5 mL were diluted in 2 Liter volumetric flask) for cations analysis, (sodium carbonate anhydrous ACS reagent, Sigma-Aldrich 0.764 gram were prepared in 2 Liter volumetric flask) for anions analysis, (Phosphoric acid  $\geq 85$  wt% solution in water, ACS reagent, Sigma-Aldrich 7mL were diluted in 1 Liter volumetric flask). During ion chromatographic separation, a cation-exchange column containing PBMA-coated silica particles was employed. The negatively charged carboxylic acid groups on the PBMA resin facilitate ion exchange interactions with positively charged cations. Due to its small ionic radius and monovalent charge, lithium exhibits a shorter retention time compared to monovalent and divalent ions such as sodium ( $\text{Na}^+$ ), potassium ( $\text{K}^+$ ), calcium ( $\text{Ca}^{2+}$ ) and magnesium ( $\text{Mg}^{2+}$ ), which interact more strongly with the stationary phase. Chromatographic data were processed by conductivity detectors, where retention times and peak areas were used to identify and quantify lithium alongside major cations like sodium and calcium. The clear separation and early elution of lithium allowed for precise detection, even within the complex ionic matrix of produced water.



**Table 2—Chemicals and Salts used in this work.**

| Chemical                                  | CAS #                                       | Manufacturer                     | MW (g/mol) | Type (Solid/Liquid) |
|-------------------------------------------|---------------------------------------------|----------------------------------|------------|---------------------|
| Anion Standard                            | 7722-76-1,7681-49-4,<br>7783-20-2,7732-18-5 | Ricca Chemical Company           | -          | Liquid              |
| Cation Standard                           | 7790-69-4,7786-30-<br>3,7732-18-5           | Ricca Chemical Company           | -          | Liquid              |
| Lithium Chloride                          | 7447-41-8                                   | Carlo Erba Reagents              | 42.39      | Solid               |
| Sodium Chloride                           | 7647-14-5                                   | VWR Chemicals                    | 58.44      | Solid               |
| Potassium Chloride                        | 7447-40-7                                   | Sigma-Aldrich                    | 74.55      | Solid               |
| Sodium Carbonate<br>Anhydrous             | 497-19-8                                    | Sigma-Aldrich                    | 105.99     | Solid               |
| Sodium hydrogen carbonate                 | 144-55-8                                    | VWR Chemicals                    | 84.01      | Solid               |
| Nitric Acid                               | 7697-37-2                                   | Sigma-Aldrich                    | 63.01      | Liquid              |
| Phosphoric Acid                           | 7664-38-2                                   | Sigma-Aldrich                    | 98         | Liquid              |
| Magnesium Chloride<br>Hexahydrate         | 7791-18-6                                   | VWR Chemicals                    | 203.30     | Solid               |
| Calcium Chloride Dihydrate                | 10035-04-8                                  | Fisher Scientific                | 147.01     | Solid               |
| Barium chloride dihydrate                 | 10326-27-9                                  | Sigma Aldrich                    | 244.26     | Solid               |
| Sodium bromide                            | 7647-15-6                                   | Alfa Aesar                       | 102.89     | Solid               |
| Sodium Iodide                             |                                             |                                  |            | Solid               |
| Strontium Chloride<br>Hexahydrate Crystal | 10025-70-4                                  | Avantor Performance<br>Materials | 266.620    | Solid               |

## Results and Discussion

### Retention Time and Elution of Lithium through PBMA Columns

The cationic components in the solution after diluting the samples 1000 times will be separated in the PBMA column based on the retention time of each ion which depends on their physiochemical properties as well as their interactions with the functional groups in the stationary phase. The IC conductivity detector used to identify ions by measuring changes in the electrical conductivity of the mobile phase as it flows through a cell with two electrodes. When ions are present, they alter the conductivity, allowing for quantification. The process involves applying a voltage to measure conductivity, separating ions based on their interaction with a stationary phase, and often using a suppressor as practical solution for the IC to enhance sensitivity by reducing background conductivity, decreases the background conductivity of the eluent, minimizes baseline noise and optimizes the signal-to-noise ratio. The detector uses digital signal processing to ensure accurate data and is designed to maintain temperature stability and a wide working range, making it a reliable and efficient tool for ion analysis ensuring precise measurement of the conductivity during the ion chromatographic determination of cations and anions. Lithium ions exhibit unique chemical properties that influence their behavior in IC. Due to their small ionic radius and high charge density, lithium ions interact has the least retention time with the stationary phase in cation-exchange leading to shorter retention times compared other alkali alkaline earth metal ions like sodium, potassium, cesium, magnesium, calcium, strontium and barium. The lithium retention time is between 6-6.9 minutes (Figure 3). Additionally, the lithium high hydration energy plays an important part of the ion-column interactions (Choi et al. 2025).

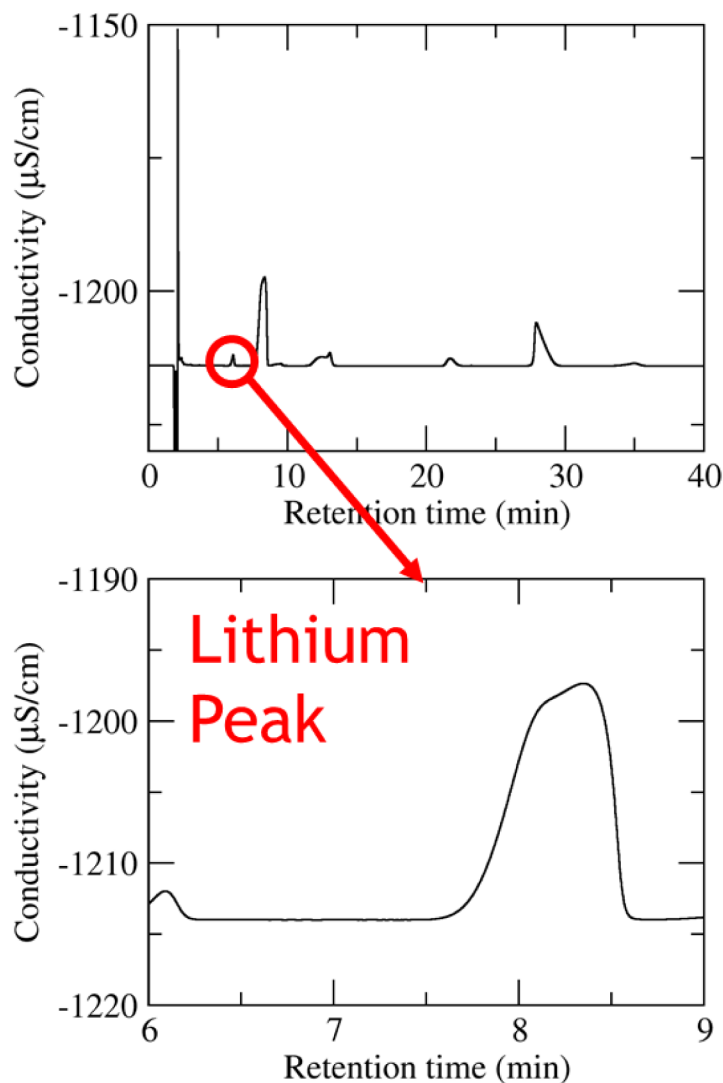


Figure 3—IC chromatogram showing lithium peak and retention time.

### Calibration of Lithium using Standard Solutions

Before conducting the analysis with the synthetic PW samples, the integrated conductivity peaks need to be compared against already known standard samples in order to quantitatively determine the concentration of lithium. The standard solution containing lithium ions is listed in [Table 2](#). We prepared six samples from the standard solution through titration using highly pure DI water as described in the methods section. The points are 0.05, 0.15, 0.1, 0.2, 1, 5, and 10 mg/L which were used to construct the calibration curve as shown in [Figure 4](#). The calibration resulted in an excellent linear curve yielding a linear regression coefficient of 0.99998 with a relative standard deviation of 0.583%.

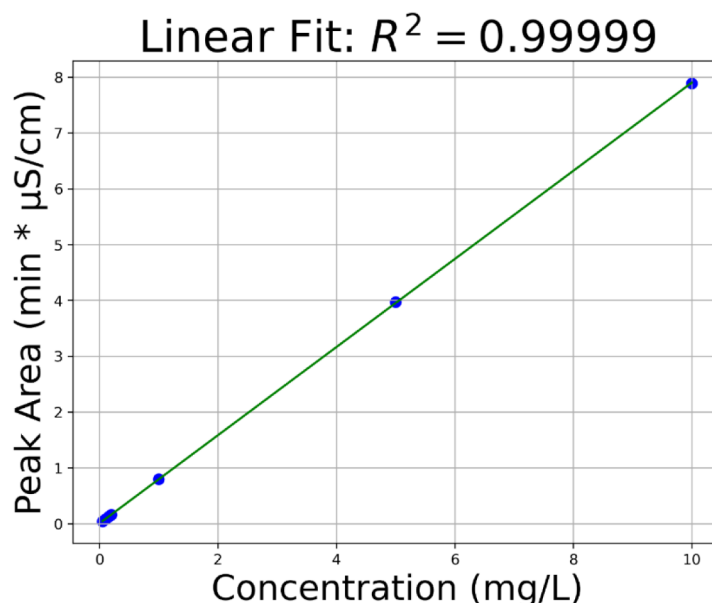


Figure 4—IC calibration curve of lithium using PBMA column and conductivity detection.

### IC System Performance for Trace Lithium Measurements in high TDS Water

Seven synthetic PW samples were prepared in order to test the performance of the IC system with the PBMA column. The seven samples and their compositions are listed in Table 3. These concentrations were chosen to represent a typical geochemical composition of subsurface brines with TDS exceeding 220,000 ppm with varying lithium content between 10 and 200 ppm. The samples were prepared as previously mentioned in the methods section. After integrating the peaks in the chromatogram for all of the seven samples, we compared the actual concentrations of lithium against those determined by the IC system. The correlation coefficient obtained from the measured brine samples was 0.9983, suggesting that the method effectively quantified lithium with limited matrix interference. Therefore, the PBMA column with conductivity column offer a simple and comparatively low-cost method for measuring the concentration of lithium in complicated brine solutions. This does not only make the IC a promising laboratory solution for quantifying subsurface brine resources of lithium, but also a mobile system that can perform the measurements in the field or as a semi-in line tool. This is possible with the IC due to the low requirements in terms of chemicals and consumables compared to the ICP systems that require significant amounts of argon and are very difficult to mobilize.

Table 3—Composition of the studied synthetic brine samples.

| Brine Number | Li <sup>+</sup> (mg/L) | Na <sup>+</sup> (mg/L) | K <sup>+</sup> (mg/L) | Mg <sup>2+</sup> (mg/L) | Ca <sup>2+</sup> (mg/L) | Sr <sup>2+</sup> (mg/L) | Ba <sup>2+</sup> (mg/L) | Cl <sup>-</sup> (mg/L) | Br <sup>-</sup> (mg/L) | I <sup>-</sup> (mg/L) | TDS (mg/L) |
|--------------|------------------------|------------------------|-----------------------|-------------------------|-------------------------|-------------------------|-------------------------|------------------------|------------------------|-----------------------|------------|
| 1            | 10                     | 20000                  | 1500                  | 5000                    | 50000                   | 5000                    | 200                     | 139399                 | 100                    | 10                    | 221219     |
| 2            | 20                     | 20000                  | 1500                  | 5000                    | 50000                   | 5000                    | 200                     | 139425                 | 150                    | 20                    | 221315     |
| 3            | 30                     | 20000                  | 1500                  | 5000                    | 50000                   | 5000                    | 200                     | 139451                 | 200                    | 30                    | 221411     |
| 4            | 50                     | 20000                  | 1500                  | 5000                    | 50000                   | 5000                    | 200                     | 139504                 | 300                    | 50                    | 221604     |
| 5            | 75                     | 20000                  | 1500                  | 5000                    | 50000                   | 5000                    | 200                     | 139536                 | 500                    | 75                    | 221886     |
| 6            | 100                    | 20000                  | 1500                  | 5000                    | 50000                   | 5000                    | 200                     | 139545                 | 750                    | 100                   | 222195     |
| 7            | 200                    | 20000                  | 1500                  | 5000                    | 50000                   | 5000                    | 200                     | 139695                 | 1500                   | 200                   | 223295     |



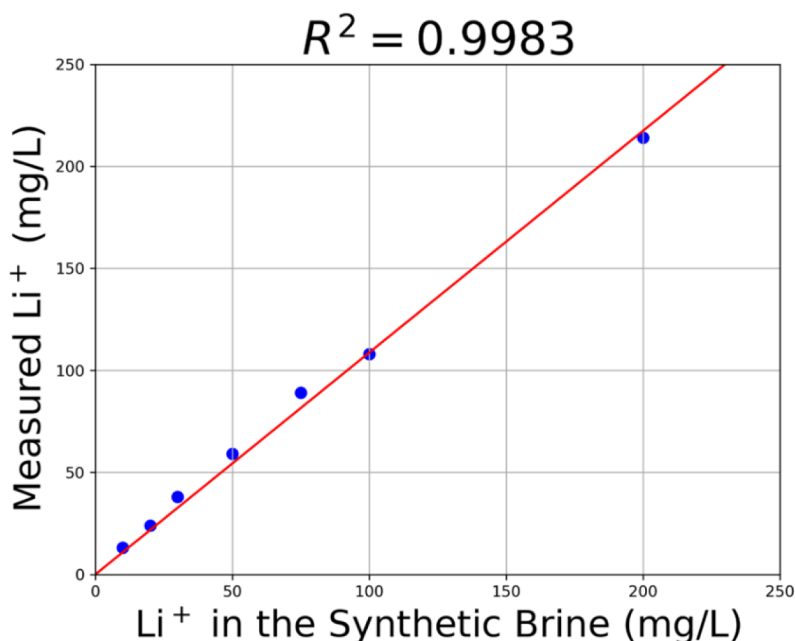


Figure 5—Measured lithium ion concentration versus the actual values in high TDS produced water matrix.

## Conclusion

In conclusion, lithium is a critical metal of paramount importance, and its extraction from unconventional sources, such as produced water, presents a promising opportunity to supplement traditional supply chains. However, the accurate measurement of lithium in these complex matrices is hindered by the high TDS and the presence of contaminants, including hydrogen sulfide and dispersed hydrocarbons. This study has demonstrated the efficacy of IC with a PBMA column and conductivity detection for lithium measurement, yielding an excellent calibration correlation coefficient of 0.99998 and a relative standard deviation of 0.583%. The robustness of this IC system was further validated through its successful analysis of high-TDS synthetic water samples containing a diverse array of cations, including sodium, potassium, magnesium, calcium, strontium, and barium, which are commonly encountered in natural geological brine samples. Notably, the results exhibited excellent agreement between actual and measured lithium concentrations, despite the complex matrix effects. These findings underscore the potential of the IC system for not only laboratory-based measurements but also for mobile, on-site applications, such as near-wellsite detection of lithium in oilfield brines, thereby advancing the efficient and effective exploration and quantification of this critical metal from produced water sources.

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